ALERT meeting

AHDC alignment

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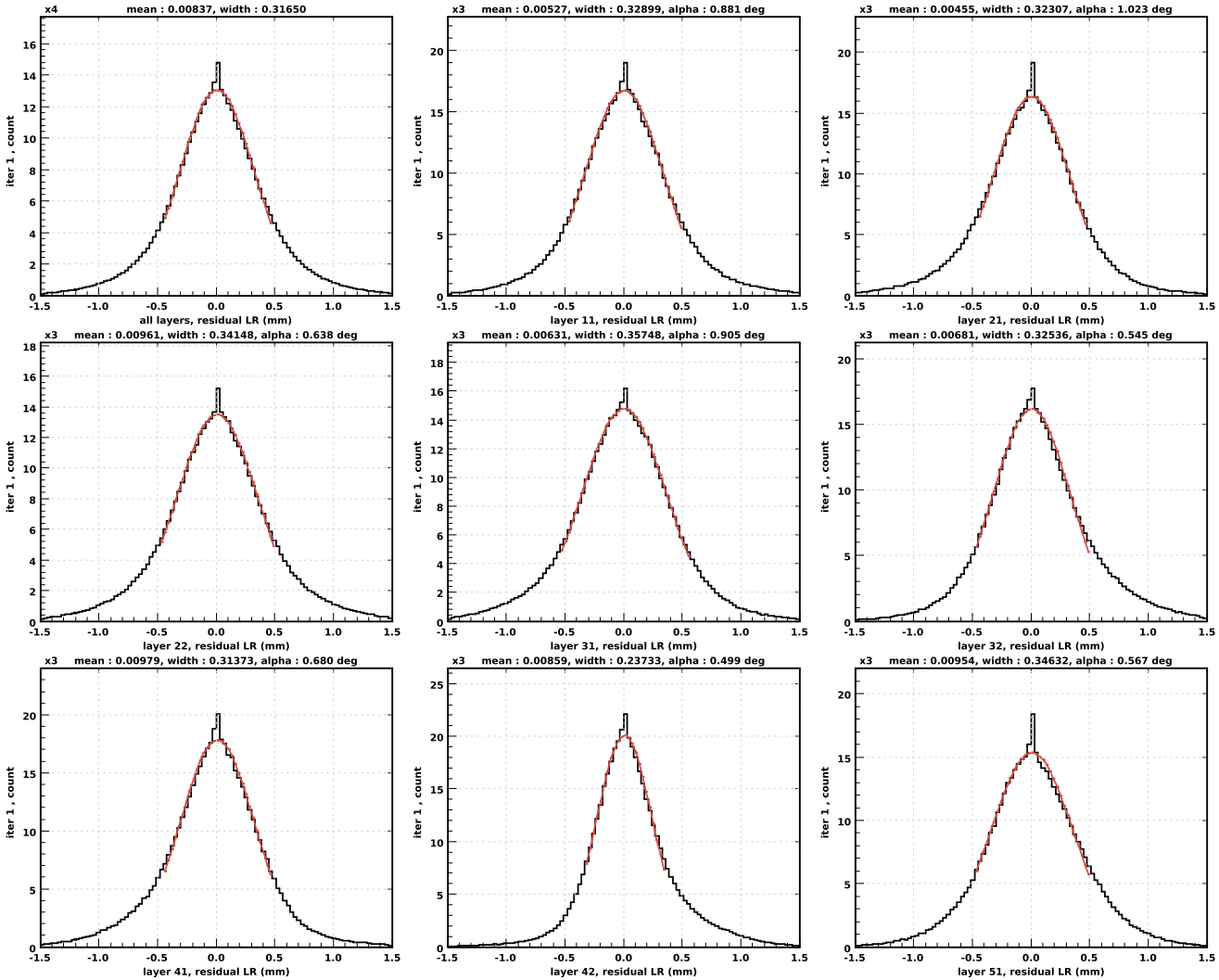
April 13, 2026

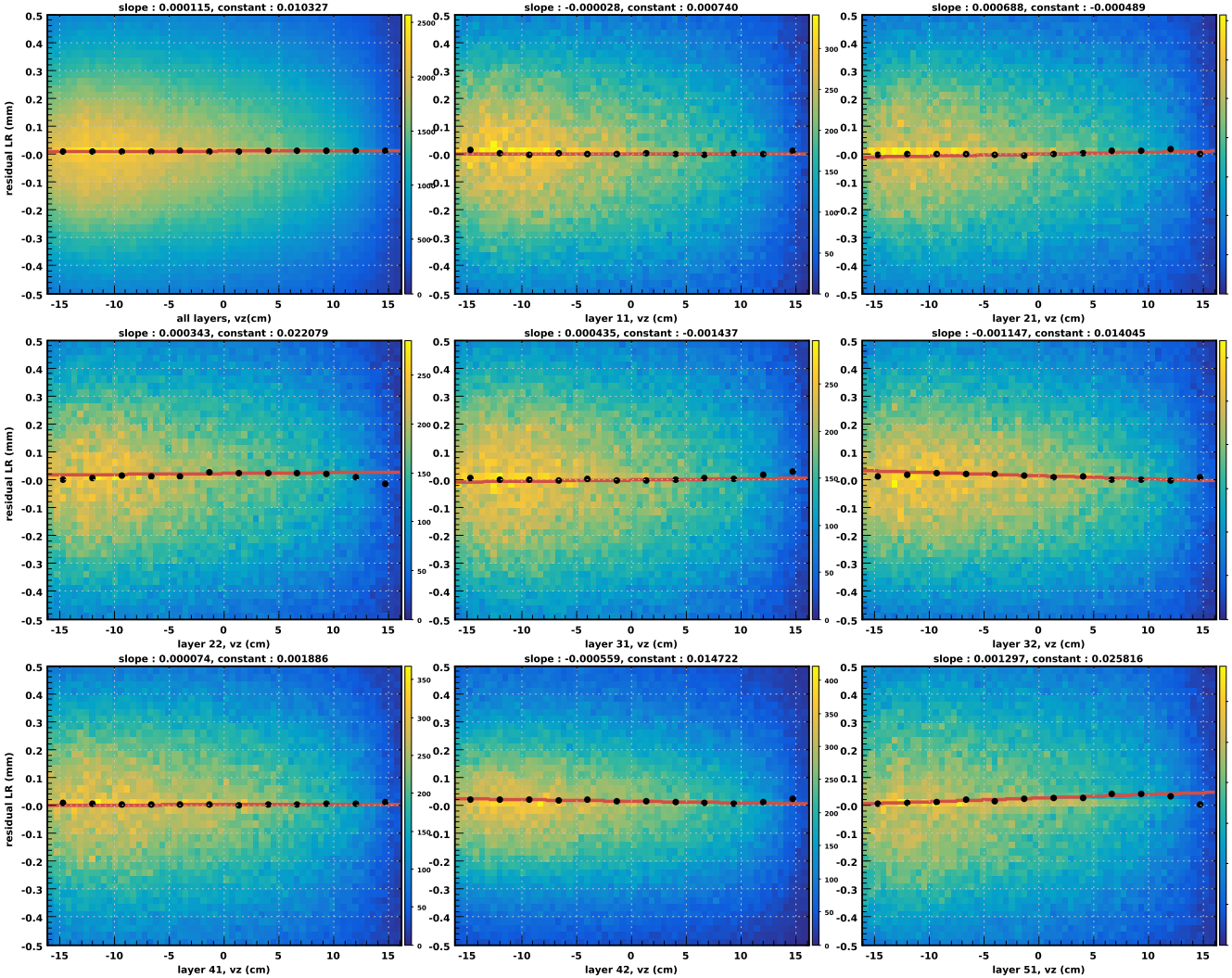
Updates - layer alignment

- I finished the main structure of the alignment code
- Current status (with KF fit): all residual peaks are below 0.01 mm.



Updates - layer alignment

- For now, the rotation is the same for the two end points of the wires
- However, we have a correlation between the residual LR and the vz
- This let think that we should rotate the two end points independently



Updates - vz alignment

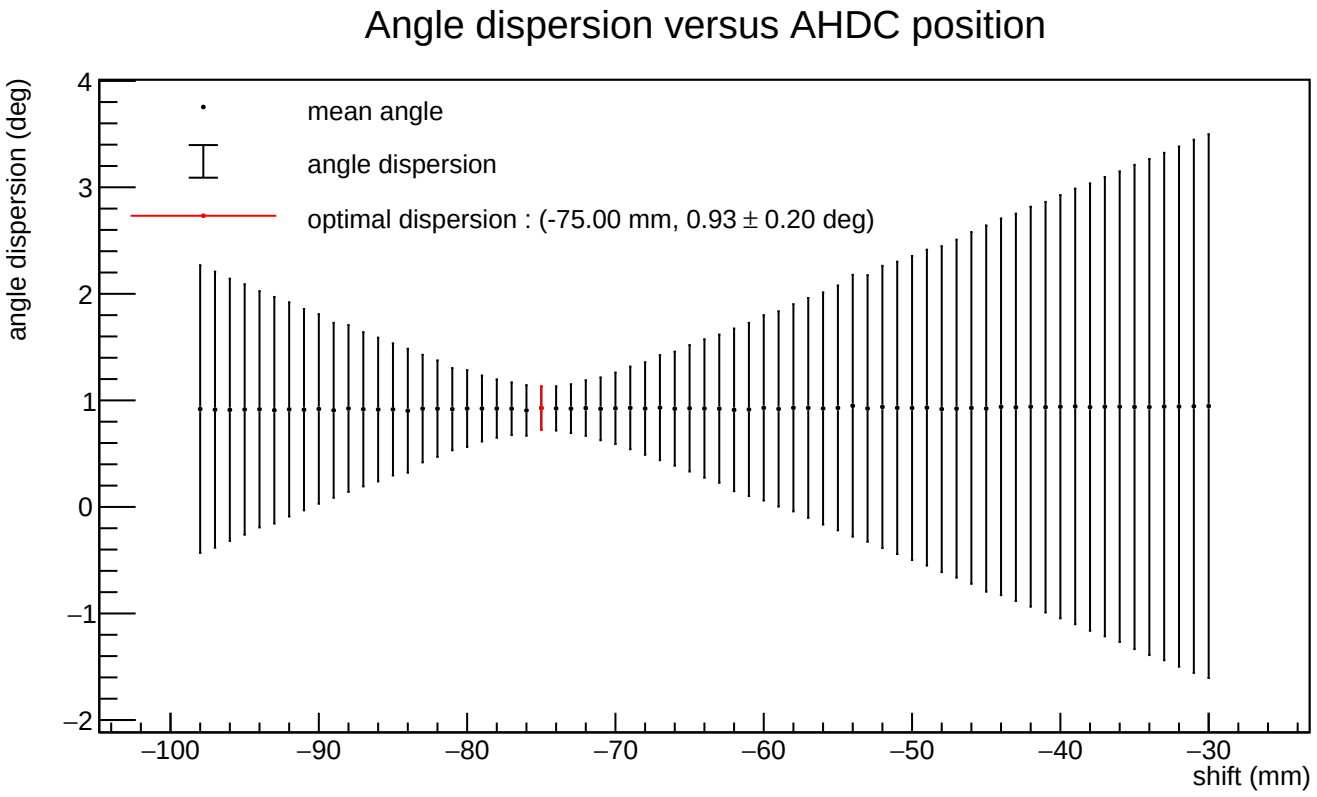
- We still need to determine the exact position of the AHDC with respect to CLAS
- Doing the alignment without KF fit, I noticed that the dispersion between the correction angles is more or less big depending on the AHDC position
- The values at vz = -70 mm are the most consistent here

AHDC position	-54 mm [1]	-30 mm	-70 mm
layer 11	0	-1.49	+0.756
layer 21	+2.55	+3.82	+1.517
layer 22	+2.11	+3.24	+0.984
layer 31	0	-1.54	+0.807
layer 32	-0.32	-1.96	+0.382
layer 41	+3.0	+3.73	+1.305
layer 42	+2.13	+3.3	+0.975
layer 51	-0.21	-1.61	+0.679

[1] hand made

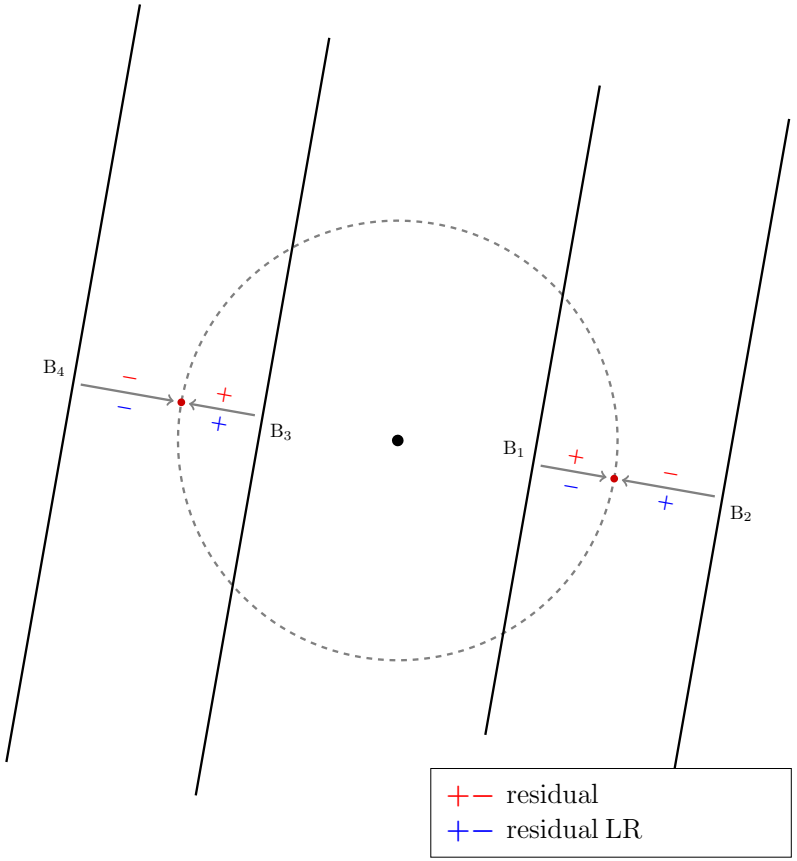
Updates - AHDC position scan

- I did AHDC position scan
- Given this study, the optimal position is -75 mm. That's what I use for the alignment



Next steps

- I will continue my work on the alignment
  - Rotate independently the two end points of the wires in a given layer
  - Push the study further by doing a wire by wire alignment
- Pending task
  - Push my work on the ATOF hits for the Kalman Filter in coatjava
  - Finish the work on the Kalman Filter (material verification, target gas density, PID, magnetic field, optimization)

Backup slides

- Procedure:
  1. We looked at elastic events : electron + track (here: collection of hits only)
  2. We computed the theoretical momentum vector of the track $\vec{p}$
  3. We made the propagation without doing any fit (i.e no KF correction)
  4. We looked at residual_LR (left-right) $\longrightarrow$ sensitive to the alignment